



Morphology III

Merlin Balkhash
Department of Linguistics, Rutgers University
2026/09/16 (updated: 2026-09-12)

Please pick up your **name card** from the front

Homophones and Allomorphs

homophones

- sound the same, but mean different things
 - *small-er* → -er = 'more' *help-er* → -er = 'agent'
 - *paper-s* → -s = PLURAL *see-s* → -s = 3SG PRESENT
- these are **different morphemes** that happen to sound the same
- **homophonous** not **homophobic**

allomorphs

- mean the same thing, but are pronounced differently
 - *wug-s*, *peff-s*, *gutch-es* *talk-ed*, *buzz-ed*, *land-ed*
 - *magic-al*, *magic-ian* *clear-ing*, *clar-ify*
- each of these is a **separate morpheme** with multiple pronunciations and each pronunciation is called an **allomorph** of that morpheme

Types of Morphemes I: Free vs. Bound

compare two **lexical entries** we have seen:

label	'dog'	'teach'
phonological representation	/dag/	/tɪ:tʃ/
semantic representation	{x: x is a dog }	{<x, y>: x teaches y}

label	'-er'	'-able'
phonological representation	/ər/	/əbəl/
semantic representation	one who does X action	able to be X -ed

free morpheme

can stand alone as a word
e.g. *dog, cat, house,*

bound morpheme

cannot stand alone
e.g. *-er, -s, cran-*

prefix-

-infix-

-suffix



circum-

-fix

Types of Morpheme II: Root vs. Affixes

root

- **"base"** morpheme to which other morphemes are attached
- contributes the **core meaning** of the word
- if a word has a single morpheme, that morpheme is a root
- examples: *danc-ing*, *dis-appear*, *help-less-ness*, *elephant*

affix

- morpheme that is **attached to the root**
- **prefixes** go before root, **suffixes** go after root – both are affixes
- a word can have multiple prefixes or suffixes
- Examples: *in-complete*, *act-ive*, *un-trust-worthy*, *will-ing-ly*

Types of Morphemes III

Inflectional vs. Derivational

inflectional

- **inflectional morphemes** are used to **inflect** words
- they add grammatical nuance but do not change the core meaning
- in English, they're mostly used to conjugate verbs (*jump* → *jump-ed*), pluralize nouns (*cat* → *cat-s*), and form the comparative and superlative of adjectives (*bright* → *bright-er* → *bright-est*)

derivational

- **derivational morphemes** are used to derive **new words**
- they change the core meaning of the word in some way
- they can, but do not always, change a word's part of speech
 - *simple* → *simpl-ify*, *exist* → *exist-ence*, *pack* → *un-pack*

Exercise I: Morphemes Types

for each word:

- indicate whether it is **mono-** or **poly**morphemic
- indicate whether it is **simple** or **compound**
- divide it into morphemes (use **hyphens**)

for each morpheme:

- indicate whether it is a **root**, **prefix**, or, **suffix**
- indicate whether it is a **free** or **bound** morpheme
- if it is an affix, indicate whether it is **inflectional** or **derivational**

1. unpacks

2. railroad

3. chameleon

4. backpacker

5. greater

6. finger

7. expectant

8. unbelievably

9. uncouth

10. meaningful

Exercise I: Morphemes Types

for each word:

- indicate whether it is **mono-** or **poly**morphemic
- indicate whether it is **simple** or **compound**
- divide it into morphemes (use **hyphens**)

for each morpheme:

- indicate whether it is a **root**, **prefix**, or, **suffix**
- indicate whether it is a **free** or **bound** morpheme
- if it is an affix, indicate whether it is **inflectional** or **derivational**

1. unpacks

un-pack-s

2. railroad

rail-road

3. chameleon

chameleon

4. backpacker

back-pack-er

5. greater

great-er

6. finger

finger

7. expectant

expect-ant

8. unbelievably

un-believ-ab-ly

9. uncouth

un-couth

10. meaningful

mean-ing-ful

Structures in Words

Why Order Matters

consider words below **morphemes** make them up

- unusable
- unlockable

Why Order Matters

consider words below **morphemes** make them up

- unusable
 - unlockable
- **un-us(e)-able**
 - **un-lock-able**

Why Order Matters

consider words below **morphemes** make them up

- unusable
- unlockable
- **un-us(e)-able**
- **un-lock-able**

what are possible meanings of these two adjectives?

- **un-us-able** has **only one meaning**
 - not able to be used
- **un-lock-able** is on the other hand, **ambiguous**
 - not able to be locked
 - able to be unlocked

Semantic Gap

Why Order Matters

consider words below **morphemes** make them up

- unusable
- unlockable
- **un-us(e)-able**
- **un-lock-able**

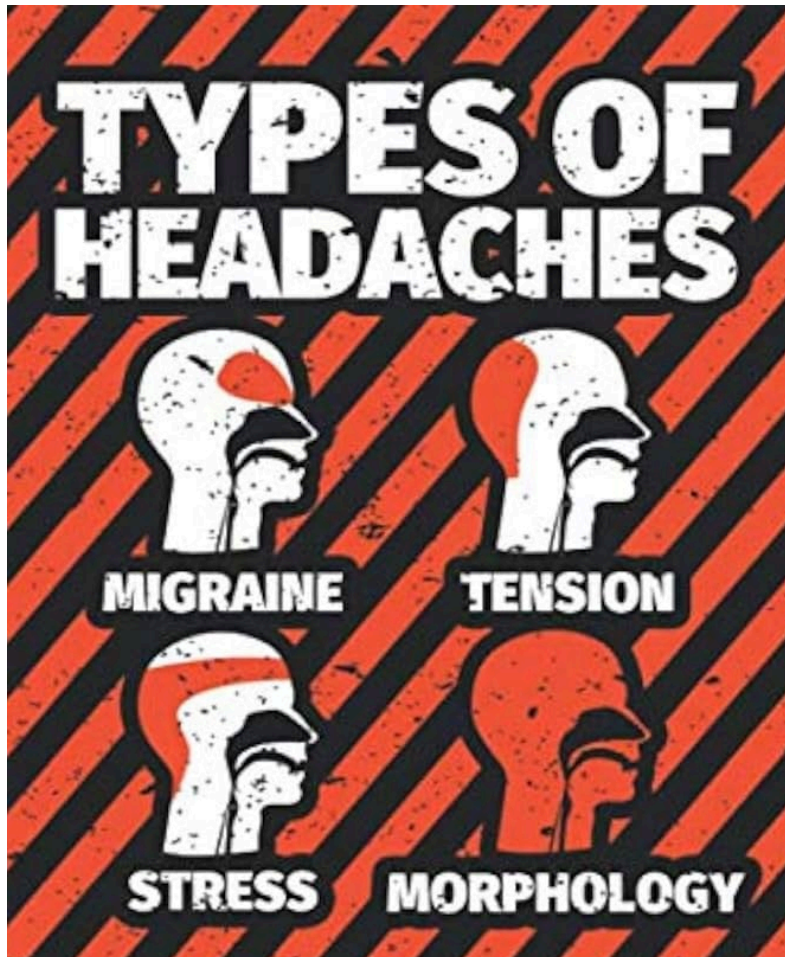
what are possible meanings of these two adjectives?

- **un-us-able** has **only one meaning**
 - not able to be used
- **un-lock-able** is on the other hand, **ambiguous**
 - not able to be locked
 - able to be unlocked

Semantic Gap

why are they different though both are formed by attaching the same morphemes to the root?

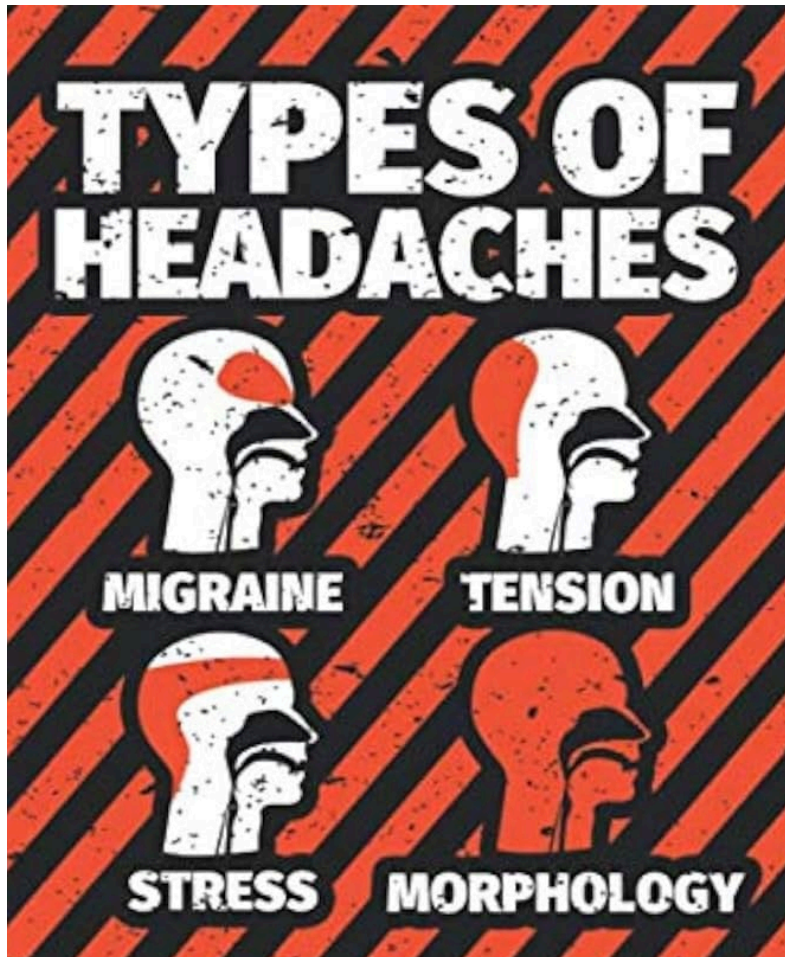
Affixes are 'Picky'



affixes have **selectional requirements**, meaning that they can only attach to stems of **a particular category**:

- **inflectional** affix:
 - comparative **-er** attaches only to adjectives
tall-er, * *cat-er*
 - plural **-s** attaches only to nouns
cat-s, * *tall-s*
- **derivational** affix:
 - **re-** attaches only to verbs
re-do, *re-visit*, * *re-red*, * *re-pencil*

Affixes are 'Picky'



- in addition to being picky about their base, **derivational** affixes also **determine** the category of the resulting word:
 - derivational **-ness** attaches to adjectives to create nouns
loneli-ness, happi-ness, concrete-ness
 - derivational **-able** attaches to verbs to create adjectives
search-able, plan-able, do-able
- affixes can be even pickier, only attaching to certain types of nouns
 - the derivational affix **-ion** attaches to verbs of only **Latin** origin to create nouns
 - *convers-ion*

Exercise IV: Adapted from *Language Files*

from the examples given for each of the following **suffixes**, determine:

1. the lexical category (i.e. parts of speech) of the **root**
2. the lexical category (i.e. parts of speech) of the word resulting from the addition of the **suffix**

a. -ive: repressive, active, disruptive, explosive

-ive takes ____, yields ____

b. -ion: invention, narration, expression, pollution

-ion takes ____, yields ____

c. re-: reelect, resubmit, restart, recap, reactivate

re- takes ____, yields ____

Exercise IV: Adapted from *Language Files*

from the examples given for each of the following **suffixes**, determine:

1. the lexical category (i.e. parts of speech) of the **root**
2. the lexical category (i.e. parts of speech) of the word resulting from the addition of the **suffix**

a. -ive: repressive, active, disruptive, explosive

-ive takes verb, yields adjective

b. -ion: invention, narration, expression, pollution

-ion takes verb, yields noun

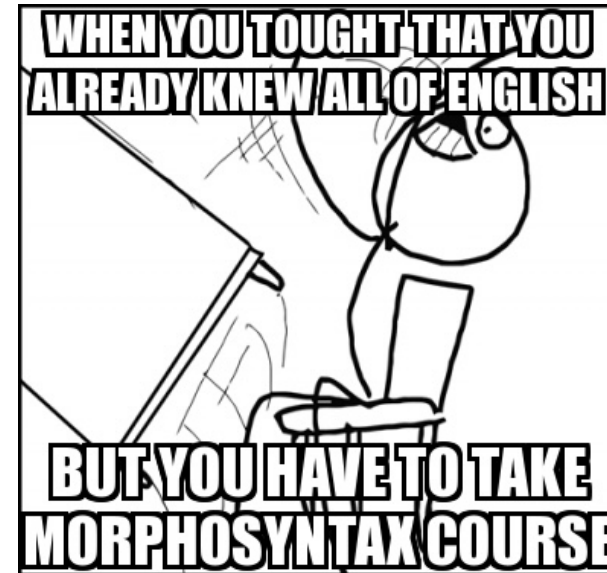
c. re-: reelect, resubmit, restart, recap, reactivate

re- takes verb, yields verb

Back to Order of Affixation

the **selectional restrictions** of affixes often help us **determine** the order in which they were attached

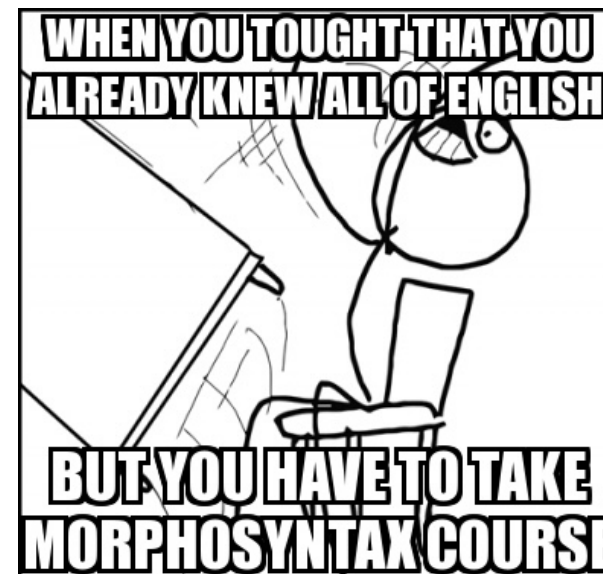
- consider the following polymorphemic word
 - **re-play-able**
- which affix was attached **first** to the root *play*?



Back to Order of Affixation

the **selectional restrictions** of affixes often help us **determine** the order in which they were attached

- consider the following polymorphemic word
 - **re-play-able**
- which affix was attached **first** to the root *play*?



- **re-** has to be attached first, since it cannot attach to adjectives
 - *play* → **re-play** → **re-play-able**

Morphological Structure

happy + -ly = happily

crazy + -ly = crazily

bright + -ly = brightly

un- + do = undo

un- + real = unreal

Morphological Structure

happy + -ly = happily

crazy + -ly = crazily

bright + -ly = brightly

adjective + -ly = adverb

un- + do = undo

un- + real = unreal

un- + verb = verb

un- + adjective = adjective

Morphological Structure

happy + -ly = happily

crazy + -ly = crazily

bright + -ly = brightly

adjective + -ly = adverb

unknowingly

un- + do = undo

un- + real = unreal

un- + verb = verb

un- + adjective = adjective

unknowingly

Morphological Structure

adjective + -ly = adverb

un- + verb = verb

un- + adjective = adjective

unknowing = ?

knowingly = ?

unknowingly = ?

unknowingly = ?

unknowingly

unknowingly

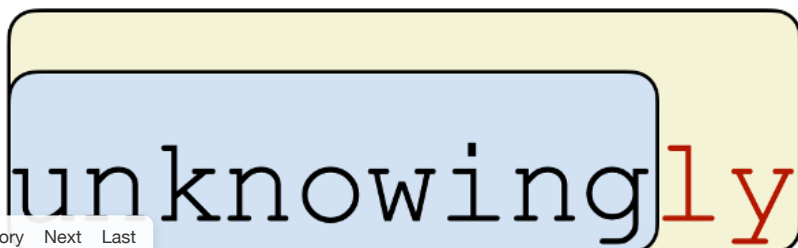
Morphological Structure

adjective + -ly = adverb

unknowing = adjective

unknowingly = adverb

YES



A diagram illustrating the correct morphological structure of the word 'unknowingly'. It consists of a large yellow rounded rectangle containing a smaller blue rounded rectangle. Inside the blue rectangle, the word 'unknowing' is written in black. To the right of 'unknowing', the letters 'ly' are written in red, positioned outside the blue rectangle but within the yellow rectangle.

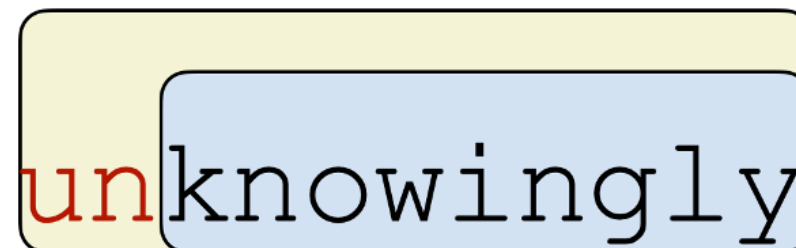
un- + verb = verb

un- + adjective = adjective

knowingly = adverb

unknowingly = adjective

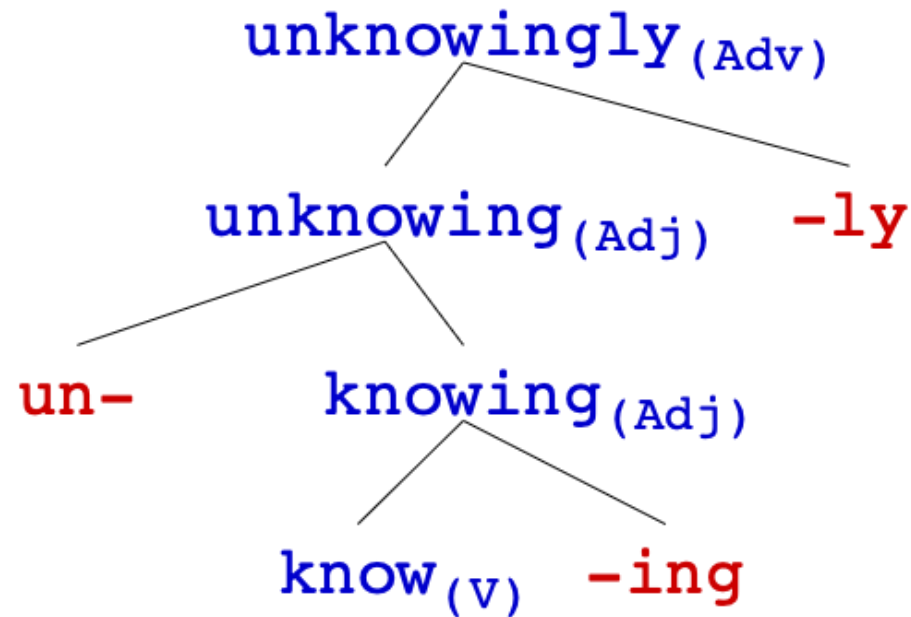
NO, wrong transformation



A diagram illustrating an incorrect morphological structure for the word 'unknowingly'. It consists of a large yellow rounded rectangle containing a smaller blue rounded rectangle. Inside the blue rectangle, the word 'knowingly' is written in black. To the left of 'knowingly', the letters 'un' are written in red, positioned outside the blue rectangle but within the yellow rectangle.

Morphological Structure

decomposing *unknowingly*



Morphological Structure



- as we said, **order of affixation** affects the **meaning** of a word
- the fact that we can't add affixes in any order suggests that words have a **hierarchical structure**
- we can visualize this structure with a **tree diagram** as above
- or an alternative is **labeled bracketing**, which is more compact but perhaps less intuitive:

[Adj un- [Adj [v lock] -able]]

[Adj [v un- [v lock]] -able]

Morphological Tree Drawing

de-humid-ify-er

1. identify the word's morphemes and write them out separately in one row
2. identify the root (or roots) and label its **part of speech**
3. look at the morphemes immediately before and after the root
check **which one is added first**, draw lines upward connecting them,
and **label** the new words's part of speech
4. this new word becomes the base you'll keep building off of
repeat step 3 using this new base

Morphological Tree Drawing

de-humid-ify-er

1. identify the word's morphemes and write them out separately in one row
2. identify the root (or roots) and label its **part of speech**
3. look at the morphemes immediately before and after the root
check **which one is added first**, draw lines upward connecting them,
and **label** the new words's part of speech
4. this new word becomes the base you'll keep building off of
repeat step 3 using this new base



Morphological Tree Drawing

de-humid-ify-er

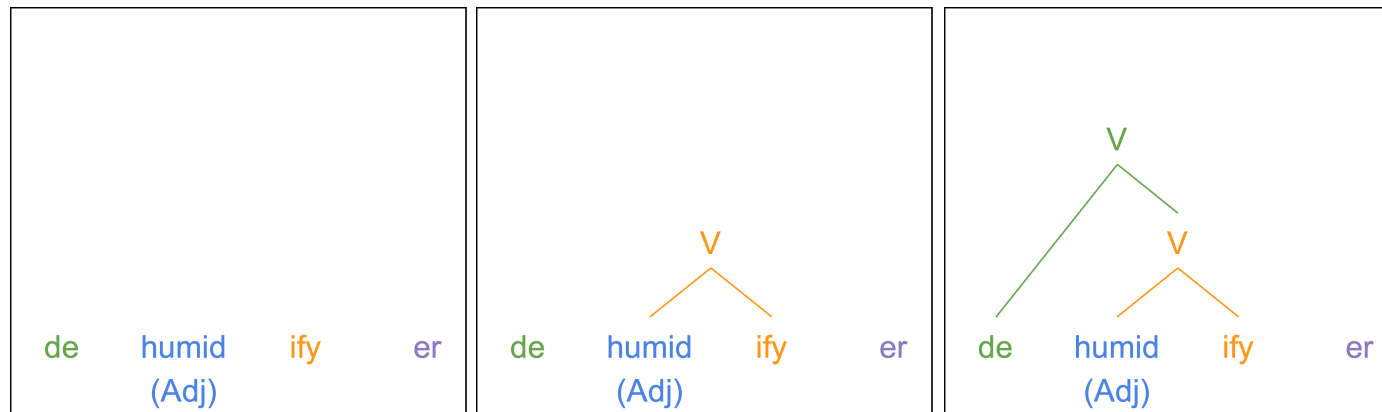
1. identify the word's morphemes and write them out separately in one row
2. identify the root (or roots) and label its **part of speech**
3. look at the morphemes immediately before and after the root
check **which one is added first**, draw lines upward connecting them,
and **label** the new words's part of speech
4. this new word becomes the base you'll keep building off of
repeat step 3 using this new base



Morphological Tree Drawing

de-humid-ify-er

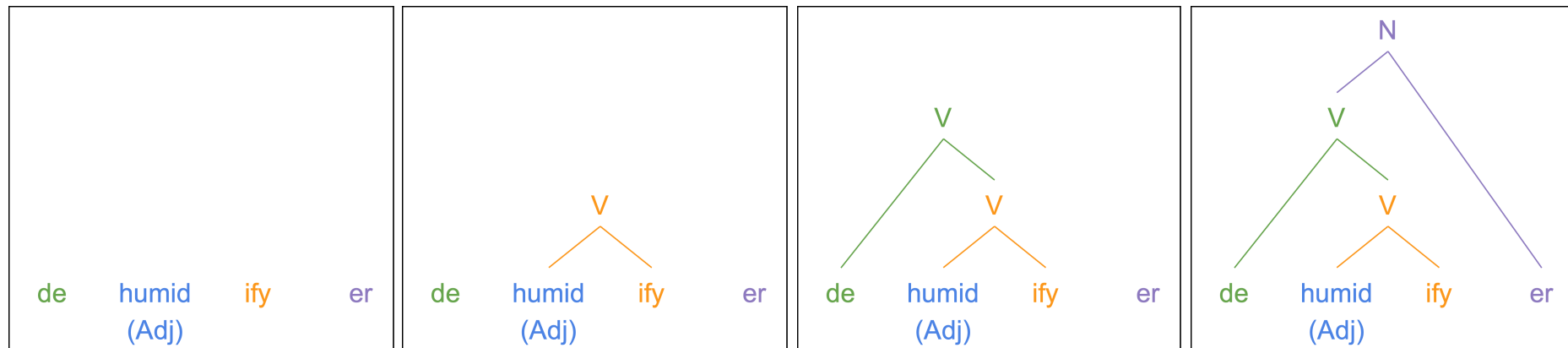
1. identify the word's morphemes and write them out separately in one row
2. identify the root (or roots) and label its **part of speech**
3. look at the morphemes immediately before and after the root
check **which one is added first**, draw lines upward connecting them,
and **label** the new words's part of speech
4. this new word becomes the base you'll keep building off of
repeat step 3 using this new base



Morphological Tree Drawing

de-humid-ify-er

1. identify the word's morphemes and write them out separately in one row
2. identify the root (or roots) and label its **part of speech**
3. look at the morphemes immediately before and after the root
check **which one is added first**, draw lines upward connecting them,
and **label** the new words's part of speech
4. this new word becomes the base you'll keep building off of
repeat step 3 using this new base



Exercise II: Morphological Structure

draw a **tree** or give **bracket labeling** for the following words:

1. disappearance
2. silliness
3. unthinkable
4. overgeneralisation
5. internationalism

HINTS

- find the root and move outward/upward
- mark the part of speech at **every** step
- if adding an affix makes a **non-word**, take one step back and try another affix instead
- if adding an affix makes it seemingly **unrelated to the overall word's meaning**, take one step back and try another affix instead



Morphological Structure: Ambiguity

verb + -able = adjective

un- + verb = verb

un- + adjective = adjective

unlock = ?

lockable = ?

unlockable = ?

unlockable = ?

unlockable

unlockable

Morphological Structure: Ambiguity

verb + -able = adjective

un- + verb = verb

un- + adjective = adjective

unlock = verb

lockable = verb

unlockable = adjective

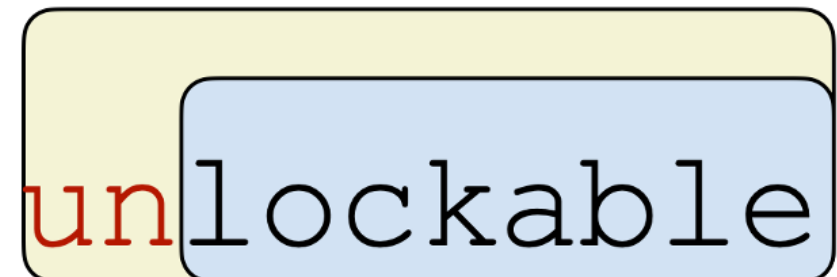
unlockable = adjective

YES

YES



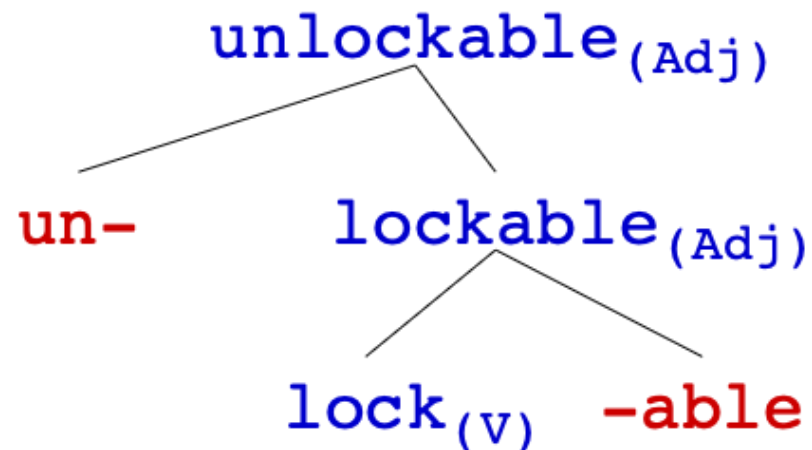
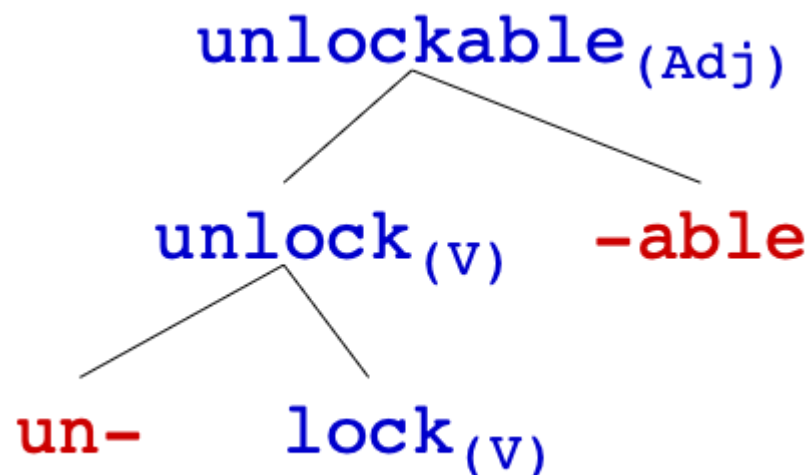
A diagram illustrating the morphological structure of the word 'unlockable'. It consists of a large yellow rounded rectangle containing a smaller blue rounded rectangle. The blue rectangle contains the text 'unlock' in black, and the yellow rectangle contains the text 'able' in red. This visualizes 'unlock' as the root verb and 'able' as the affix that turns it into an adjective.



A diagram illustrating the morphological structure of the word 'unlockable' from a different perspective. It consists of a large yellow rounded rectangle containing a smaller blue rounded rectangle. The yellow rectangle contains the text 'un' in red, and the blue rectangle contains the text 'lockable' in black. This visualizes 'un-' as the affix and 'lockable' as the root verb.

Morphological Structure: Ambiguity

tree representations can capture **different** hierarchical structures



Morphological Structure: Ambiguity

un + able



undo + able



- the case of **unlockable** is an example of a word with an **ambiguous structure**
- depending on the structure you assign it, it has a **different meaning**
- both meanings are acceptable
- which means the **meaning** of words is directly affected by their morphological structure
- that's why **structure** is important, without knowing the structure, we would not be able to be **compositional** properly

Exercise III: Structural Ambiguity

are following words ambiguous?

- if yes, provide **labeled bracketings** and **meanings** for both readings
- if no, explain why not

1. **im**measur**able**

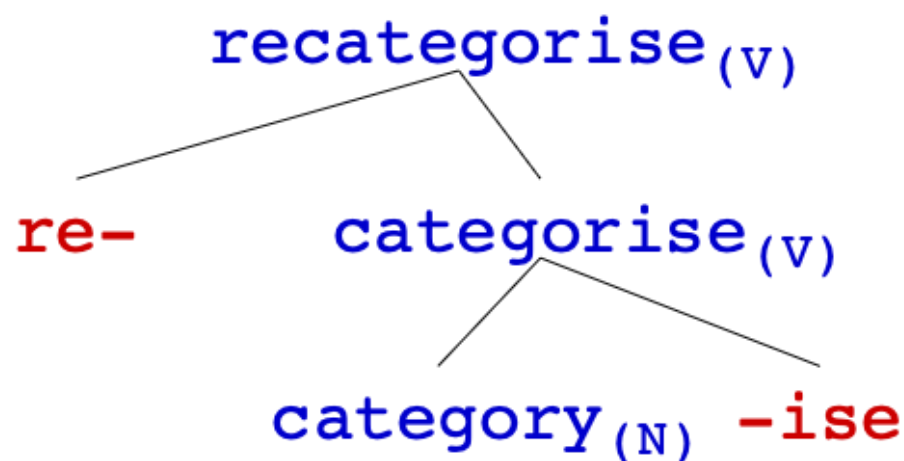
2. **un**wrapp**able**

3. **un**bear**able**

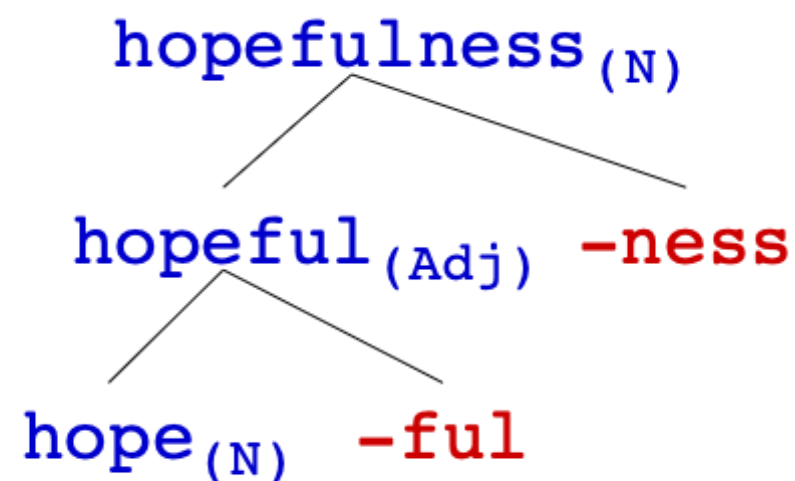


Note 1: Linear Order $\neq$ Hierarchical Order

re-category-ise



hope-ful-ness



Note 2: Ungrammaticality

sometimes neither order works: * *enjoyful*

- **-ful**: $N \rightarrow \text{Adj}$
- **en-**: $N \rightarrow V$

enjoy + ful ✗

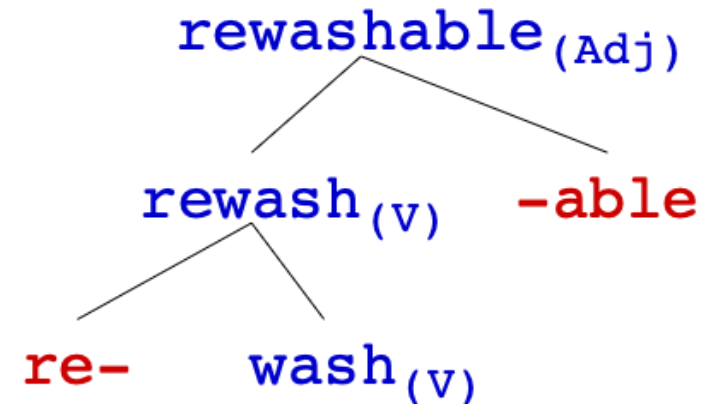
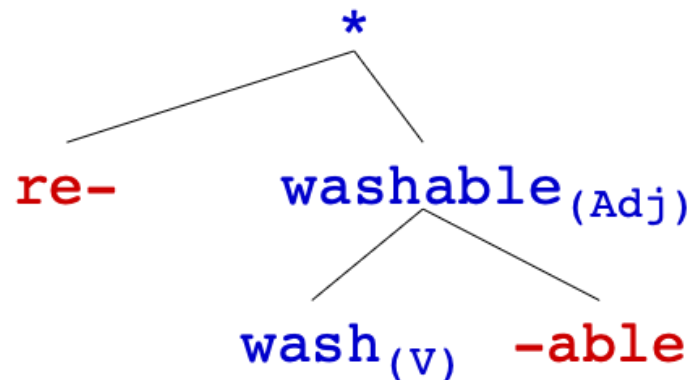
en + joyful ✗

Note 3: Comparing Analysis

sometimes both orders seem working — but not really

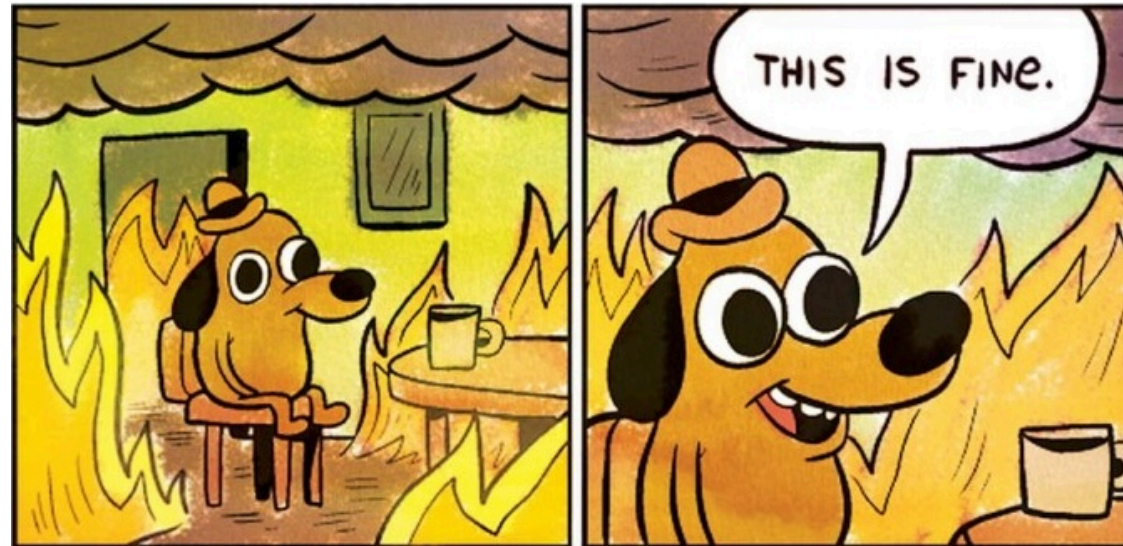
- **-able**: $V \rightarrow \text{Adj}$: wash-able, do-able
- **re-**: $V \rightarrow V$, but NOT $V \rightarrow \text{Adj}$: * re-big, * re-sad

re-wash-able



Homework I will be published this today, and it is due next Friday (**Sept 25**)

continue reading Sections 4.3, 4.4 and 4.5 of Chapter 4: Morphology from *Language Files*



Slides created via the R package **xaringan**.